Haochi Wu

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Advancing the renewable transition and climate adaptation of future energy systems through data-driven modeling and macro-energy analysis.

EMPLOYMENT

Stanford University — Stanford, CA **Sep 2025 – Present**

Department of Civil and Environmental Engineering

- Postdoctoral Researcher
- Research: industrial decarbonization, water-energy nexus

University of Michigan — Ann Arbor, MI **Jan 2025 – Sep 2025**

Center for Sustainable Systems (CSS)

School for Environment and Sustainability (SEAS)

- Research Associate (Jan 2025 – Sep 2025), Mentor: Prof. Michael Craig

Participated in the Project: Making Decarbonization of the Electric Power Sector Robust to Climate Change (NSF CAREER).

EDUCATION

University of Michigan — Ann Arbor, MI **Sep 2023 – Sep 2025**

School for Environment and Sustainability

- Visiting Ph.D. Researcher (Aug 2023 – Sep 2025)
- Supervisor: Prof. Michael Craig (U Michigan)
- Dissertation: Analysis and optimization of energy systems for climate adaptation

Zhejiang University — Hangzhou, China **Sep 2020 – Jun 2025**

School of Control Science and Engineering

- Ph.D. in Control Science and Energy Engineering (Sep 2023 – Jun 2025)
- M.Sc. in Control Science and Engineering (Sep 2020 – May 2023)
- Supervisor: Prof. Mingyang Sun (Peking University)

North China Electric Power University — Beijing, China **Sep 2016 – May 2020**

School of Control and Computer Engineering

- Bachelor of Automation Engineering with Honours
- Honors: National Scholarship (Top 1%), Rankings 4/123

SELECTED PUBLICATIONS

(Complete publication list available on Google Scholar)

- **Wu, H.**, Kong, Q., Huber, M., Craig, M., et al. (2025). Climate Change Will Increase High Temperature Risks, Degradation, and Costs of Rooftop Photovoltaics Globally. ***Joule***. (In Press).

• **Wu, H.**, Sun, M., & Craig, M. (2025). Updating Global Green Hydrogen Production Costs and Configurations under Future Climates. *The Innovation (*Cell Press*)*. (Accepted).
• **Wu, H.**, Wang, J., Teng, F., et al. (2025). Tracking Bitcoin-Induced Carbon Trajectory in China Via Refined Spatiotemporal Assessment. *Engineering*. (In Press).
• **Wu, H.**, Qiu, D., Zhang, L., & Sun, M. (2024). Adaptive Multi-Agent Reinforcement Learning for Flexible Resource Management in a Virtual Power Plant with Dynamic Participating Multi-Energy Buildings. *Applied Energy*, 374, 123998.
• Yuan, Q., **Wu, H.**, et al. (2025). PrivLoad: Privacy-preserving Load Profiles Synthesis Based on Diffusion Models. *IEEE Transactions on Smart Grid*. (Early Access).
• Meng, J., **Wu, H.**, Wang, T., et al. (2021). A novel super-cooling enhancement method for a two-stage thermoelectric cooler using integrated triangular-square current pulses. *Energy*, 217, 119360.

IN-PROGRESS PUBLICATIONS

• **Wu, H.**, Chen, J., Vaishnav, P., Craig, M. Technological improvements in EV batteries offset climate-induced durability challenges. ***Nature Climate Change***. (Under Revision).

• Moraski, J., **Wu, H.**, Craig, M., Callaway, D. Quantifying the Power System Benefits of Building Upgrades for Passive Survivability in a Changing Climate. *Environmental Science & Technology*. (Under Review).

SELECTED EXPERIENCE, AWARDS, and HONORS

2024-2025, MES Fellow, Macro Energy Systems (MES) Committee
2023-Present, Serve as Reviewer in Communications Earth & Environment, IEEE Transactions on Power Systems, IEEE Transactions on Smart Grid, IEEE Transactions on Industrial Informatics.
2021, Outstanding Graduate Student Award, Zhejiang University
2019, National Scholarship, Ministry of Education of China (Rate: top 1%)

SKILLS

• Domain Knowledge: Scientific Programming, Energy System Modeling, Power System Economics, Convex Optimization, Machine Learning, Earth and Climate Science
• Scientific Computing and AI: Python, PyTorch, Julia, Matlab/Simulink, Git, LaTeX, C, C++